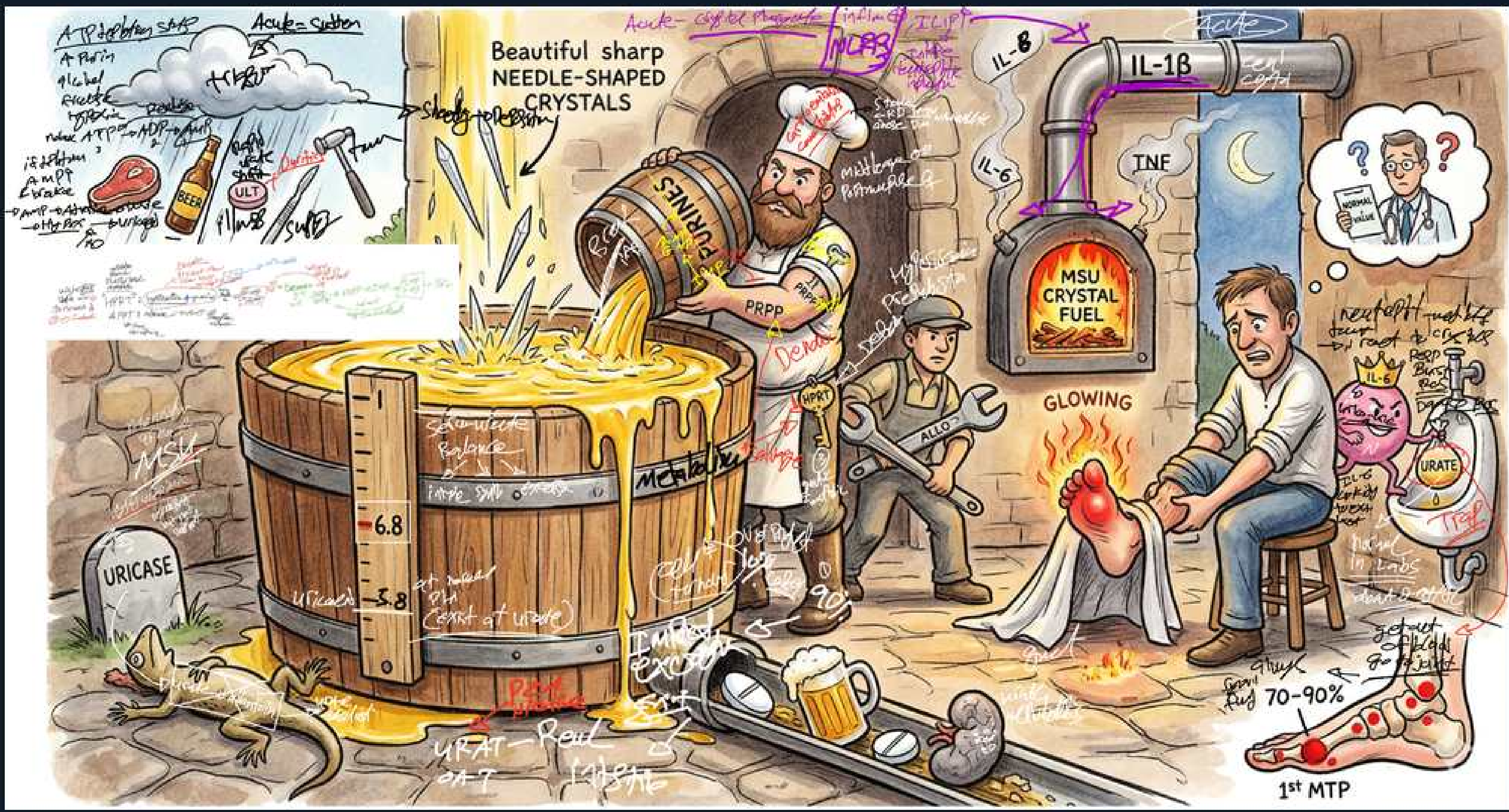
Gout: Hyperuricemia Pathophysiology & Crystal Formation



1. Needle-shaped MSU crystals

WHAT YOU SEE
'Beautiful sharp NEEDLE-shaped crystals' label top

WHAT IT ENCODES
Monosodium urate precipitates as needle-shaped crystals that are negatively birefringent under polarized light. These crystals nucleate when serum urate exceeds the solubility limit ~6.8 mg/dL.

BOARD TRAP
Rhomboid + positively birefringent = CPPD/pseudogout, the opposite crystal — do not mix up shape or birefringence.

2. Purine overproduction

WHAT YOU SEE
Top-left 'SHOP' with hammer/breakdown icons, meat, beer

WHAT IT ENCODES
Hyperuricemia from OVERPRODUCTION: high cell turnover (tumor lysis, hemolysis, psoriasis), purine-rich diet (red meat, organ meat, seafood), alcohol, and enzyme defects (HGPRT deficiency = Lesch-Nyhan, PRPP synthetase overactivity).

BOARD TRAP
Most clinical gout (~90%) is UNDER-excretion, not overproduction — overproduction is the minority cause despite being emphasized.

3. PRPP / de novo synthesis

WHAT YOU SEE
'PRPP' on the chef pouring barrel

WHAT IT ENCODES
PRPP (phosphoribosyl pyrophosphate) drives de novo purine synthesis; PRPP synthetase overactivity increases urate. Purine catabolism flows hypoxanthine → xanthine → uric acid via xanthine oxidase.

BOARD TRAP
PRPP amidotransferase is the committed/rate-limiting step — PRPP synthetase overactivity (X-linked) causes overproduction gout.

4. Xanthine oxidase (barrel pour)

WHAT YOU SEE
Chef pouring yellow uric acid from barrel

WHAT IT ENCODES
Xanthine oxidase converts hypoxanthine→xanthine→uric acid, generating urate. It is the drug target of allopurinol and febuxostat (xanthine oxidase inhibitors), which lower urate production.

BOARD TRAP
Allopurinol + azathioprine/6-MP is dangerous — blocked xanthine oxidase causes thiopurine toxicity; reduce dose or avoid.

5. Uricase (gravestone)

WHAT YOU SEE
'URICASE' gravestone with lizard

WHAT IT ENCODES
Humans LACK uricase (urate oxidase), so uric acid is our end product (unlike most mammals that convert it to soluble allantoin). Recombinant uricase (pegloticase, rasburicase) is used for refractory/tophaceous gout and tumor lysis.

BOARD TRAP
Rasburicase is contraindicated in G6PD deficiency — it generates hydrogen peroxide causing hemolysis/methemoglobinemia.

6. Under-excretion (kidney)

WHAT YOU SEE
Barrel pH markers 6.8 and 5.5, drainage

WHAT IT ENCODES
~90% of gout is renal UNDER-excretion of urate. Acidic urine (low pH) and reduced fractional excretion promote crystallization. Thiazide/loop diuretics, low-dose aspirin, cyclosporine, and CKD reduce excretion.

BOARD TRAP
Uricosurics (probenecid) work by increasing renal excretion but are contraindicated in uric-acid nephrolithiasis and low GFR.

7. MSU crystal fuel (furnace)

WHAT YOU SEE
Glowing furnace 'MSU CRYSTAL FUEL'

WHAT IT ENCODES
Deposited MSU crystals are phagocytosed and activate the NLRP3 inflammasome, releasing active IL-1β — the central driver of the gout flare and inflammatory amplification.

BOARD TRAP
The crystal itself is the danger signal; colchicine works by inhibiting microtubule-mediated crystal phagocytosis/neutrophil migration, not by lowering urate.

8. IL-1β inflammasome

WHAT YOU SEE
'IL-1B' pipe valve over furnace

WHAT IT ENCODES
NLRP3 inflammasome → caspase-1 → IL-1β is the key cytokine in acute gout. IL-1 inhibitors (anakinra, canakinumab) treat refractory flares when NSAIDs/colchicine/steroids fail or are contraindicated.

BOARD TRAP
IL-1β (not IL-1α or IL-6) is the pivotal target — canakinumab is anti–IL-1β monoclonal.

9. IL-6 / TNF amplification

WHAT YOU SEE
'IL-6' and 'TNF' pipe labels
WHAT IT ENCODES
Downstream of IL-1β, IL-6 and TNF amplify the inflammatory cascade, recruiting neutrophils and producing systemic features (fever, elevated CRP) during severe flares.
BOARD TRAP
Anti-TNF/anti–IL-6 biologics are not standard gout therapy; the gout-specific cytokine target is IL-1β.

10. Acute red hot joint

WHAT YOU SEE
Patient holding glowing red foot
WHAT IT ENCODES
Acute flare: monoarticular, intensely painful, red, hot, swollen joint peaking in 12–24h, often nocturnal. First-line: NSAIDs, colchicine, or systemic/intra-articular corticosteroids.
BOARD TRAP
Do not initiate urate-lowering therapy during an untreated acute flare without anti-inflammatory prophylaxis — it can prolong the attack.

11. First MTP (podagra)

WHAT YOU SEE
'1st MTP' foot diagram bottom right
WHAT IT ENCODES
The first metatarsophalangeal joint (podagra) is involved in ~50% of first attacks and ~70-90% of patients eventually. Cooler distal joint temperature favors urate crystallization.
BOARD TRAP
Podagra is classic but not exclusive — gout also hits midfoot, ankle, knee; isolated podagra can also be pseudogout or septic.

12. 70-90% MTP involvement

WHAT YOU SEE
'70-90%' annotation near foot
WHAT IT ENCODES
Up to 70-90% of gout patients have first-MTP involvement over the disease course. Lower-extremity predominance reflects temperature-dependent urate solubility.
BOARD TRAP
Polyarticular upper-limb gout suggests chronic tophaceous or atypical disease — early gout is typically lower-limb monoarticular.

13. Tophi deposition

WHAT YOU SEE
Pink lumpy tophus icon with crown near patient
WHAT IT ENCODES
Chronic untreated hyperuricemia deposits tophi in joints, tendons, and soft tissue, causing erosive 'rat-bite' bony lesions with overhanging edges on X-ray and joint destruction.
BOARD TRAP
Rat-bite erosions have sclerotic overhanging margins — unlike RA's marginal erosions without overhang or sclerosis.

14. Alcohol / beer trigger

WHAT YOU SEE
Beer mugs at base of vat
WHAT IT ENCODES
Alcohol — especially beer (rich in guanosine purines) — raises urate by increasing production and reducing renal excretion (lactate competes for tubular secretion). A major modifiable flare trigger.
BOARD TRAP
Wine is less gout-provoking than beer/spirits; fructose-sweetened drinks also raise urate, an underappreciated dietary trigger.

15. Diet purines (meat)

WHAT YOU SEE
Red meat and food icons upper left
WHAT IT ENCODES
Purine-rich foods (red/organ meats, anchovies, sardines, shellfish) raise urate. Dietary modification plus weight loss and reduced alcohol/fructose are first-line lifestyle measures.
BOARD TRAP
Dairy and coffee LOWER urate risk — not all dietary advice is restrictive; some foods are protective.